SANATHKUMAR DESAI

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LinkedIn | GitHub

EDUCATION

Nitte Meenakshi Institute of Technology | B.E., Electronics and Communication Engineering | Expected 2028
CGPA: 8.67/10 | Currently in 5th semester | Class XII: 95% (PCM) | Class X: 98.24%

TECHNICAL SKILLS

Programming: C; Python (basic)
Embedded & RTOS: ESP32, STM32, Arduino; FreeRTOS, Zephyr RTOS; ESP-IDF
Tools & Interfaces: STM32CubeIDE, STM32CubeMX, Arduino IDE, Git; I^2C, I^2S, PWM; UART/SPI (familiarity)
Relevant Concepts: Feedback control (PID/PD), IMU interfacing, audio sampling and filtering, motor actuation

PROJECTS

PID-Based Stationary Heading Correction Rover | ESP32, C, ESP-IDF, FreeRTOS, BNO055 | GitHub

- Developed an IMU-feedback heading controller for a four-motor rover, using I^2C yaw readings, angular-error normalization and differential PWM motor commands to correct orientation.
- Organized sensor, control and motor operations with FreeRTOS tasks; designed and assembled motor wiring, power distribution and battery/buck-converter integration, resolving electrical faults during testing.

INMP441 MEMS Microphone Interfacing & Audio Processing | ESP32, C, ESP-IDF, I^2S, UDP | GitHub

- Interfaced an INMP441 digital microphone with the ESP32 I^2S peripheral to acquire PCM audio samples and stream audio over UDP to a receiving system.
- Experimented with high-pass filtering and automatic gain control (AGC) to improve noisy raw audio output; explored buffering and real-time embedded audio processing.

INTERNSHIP EXPERIENCE

Hardware Design Intern — Mediglint LLP | Feb 2025 – Aug 2025

- Designed a medicine-dispenser CAD model in Onshape, evaluated iterative 3D-printed prototypes, and developed ESP32-controlled actuation using electromagnets and relays.
- Designed and integrated the supporting electronic architecture; final system design was submitted for a design patent.

ROS 2 Robotics Intern — NEWRRO TECH LLP | Feb 2025 – Mar 2025

- Developed an ESP32–ROS 2 serial communication bridge and a ROS 2 WebSocket-based keyboard teleoperation interface for an RC robotic car.
- Assisted in testing internal robotics libraries and documenting Arjuna Robotics Car system architecture and workflows.

LEADERSHIP & VOLUNTEERING

International Guest Management Volunteer — Ōta TOKYO 2026 | Jan 2026

- Coordinated schedules and stage movement for a Japanese guest performer at an event with approximately 15,000 attendees; worked with organizers to adapt the program to rain-related delays and facilitated audience interactions.

CERTIFICATION & LANGUAGES

Japanese-Language Proficiency Test (JLPT) N5 — Passed, July 2026 test
Languages: Kannada, English, Japanese (JLPT N5)